

ITEM: ORGANIC CHEMISTRY – ALCOHOLS, PHENOLS, CARBONYL GROUPS, ALKANES & ALKENES

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A research chemist was tasked with designing a multi-step synthesis in the laboratory. She started with but-1-ene ($\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}_3$) and carried out a series of reactions to obtain three target compounds:

Compound A, a primary alcohol obtained via hydration of the alkene

Compound B, a ketone formed from oxidation of Compound A

Compound C, a phenol obtained by further functional group transformation from Compound B

During the study, she also explored the reaction of butane ($\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_3$) under chlorination in sunlight to obtain haloalkanes, which were then used in nucleophilic substitution reactions to prepare derivatives of Compound A. The mechanisms of hydration, oxidation, and substitution were carefully observed and recorded.

Tasks

(a) Identify Compounds A, B, and C. Draw their structures.

(b) Write the equations for the following transformations:

Hydration of but-1-ene to Compound A

Oxidation of Compound A to Compound B

Conversion of Compound B to Compound C

(c) Mechanisms

*Show the mechanism of the acid-catalyzed hydration of but-1-ene to Compound A.

*Show the mechanism for the oxidation of primary alcohol to ketone.

*Show the mechanism of nucleophilic substitution of the haloalkane derived from butane to give an alcohol derivative.

(d) Explain why butane does not react with cold bromine water but reacts under sunlight.

(e) Suggest one industrial application of:

Compound B

Compound C

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He is the one who sets challenging and story-based items like this to prepare you for success.

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